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Methods and apparatus for treating glottic insufficiency

Abstract

An example implant system to treat glottic insufficiency is disclosed. The implant system includes a fixation frame including a first set of flanges at a first edge of the fixation frame and a second flange at a second edge of the fixation frame, wherein the fixation frame is configured to secure the implant system at an opening of the patient's thyroid cartilage, a port system disposed in the fixation frame and configured to receive, deliver, maintain, or remove a filler, and a flexible member, coupled to the fixation frame and in fluid or air communication with the port system, wherein based on an amount of the filler in the flexible member, the flexible member is configured to inflate in a direction to push against the patient's arytenoid cartilage so the arytenoid cartilage is rocked, adducted and/or rotated inferomedially.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is continuation application of U.S. patent application Ser. No. 15/550,033, filed Aug. 10, 2017, which is a U.S. National Stage filing under 35 U.S.C. § 371 of International Application PCT/US2017/026724, filed Apr. 7, 2017, which claims the benefit of U.S. Provisional Application No. 62/320,444, filed Apr. 8, 2016, U.S. Provisional Application No. 62/341,610, filed May 25, 2016, U.S. Provisional Application No. 62/419,972, filed Nov. 10, 2016, and U.S. Provisional Application No. 62/455,515, filed Feb. 6, 2017. The U.S. Patent Application, the International Application, and the U.S. Provisional Applications, including any appendices or attachments thereof, are incorporated by reference herein in their entirety.

BACKGROUND

(1) Dysphagia associated with aspiration pneumonia often occurs in patients with neurological disorders. The neurological disorders may be caused by stroke, brain surgery, head and/or spinal cord trauma, oropharyngeal diseases, radiation therapy, cardiac/thoracic surgery, autoimmune or other degenerative neurologic diseases. The aspiration pneumonia may be mainly caused by glottic insufficiency, due to vocal fold paralysis with or without swallowing dysfunction. Stroke patients with aspiration symptoms may have a seven-time higher risk in developing aspiration pneumonia than other types of patients. For these stroke patients, even after recovery, there is still a relatively high incidence of dysphagia associated with aspiration pneumonia.

(2) Conventional surgical techniques to treat dysphagia and glottic insufficiency may include Type I Medialization Thyroplasty (MT) procedure and Arytenoid Adduction (AA) procedure. Type I MT procedure is the main phonosurgical procedure performed in patients with glottic insufficiency. The

primary limitations of Type I MT procedure include the inability to close a wide posterior glottal chink and restore the physiological swallowing steps, like laryngeal elevation and vocal fold movement. For patients with vocal fold paralysis and a significant posterior glottic gap after the Type I MT procedure, an AA procedure may be performed subsequently to close the incompletely closed posterior glottis. Still, one limitation of the AA procedure, associated with the posterior airway closure, is an increased frequency of postoperative airway complaints after the AA procedure, due to postoperative tissue edema in the glottic area. Further, Type I MT and AA procedures may not be suitable for patients having difficulty with prolonged periods of supine positioning or intolerable for long lasting surgical procedures.

(3) Since the above two procedures either use implants or suture fixation technique, a common complaint from these procedures is the inability to precisely adjust the implant or sutures intraoperatively and postoperatively. Specifically, it is difficult to accurately perform intraoperative adjustment of implant due to edematous swelling of laryngeal mucosa caused by these procedures. For example, carving an implant during surgery may result in prolonged operation time and suboptimal shaping of the implant. Furthermore, these implants cannot be postoperatively adjusted at all.

(4) Providing an adjustable implant for the procedure would shorten the operation time, and reduce the risk of postoperative airway compromise. The size of the adjustable implant would be customized for each individual's needs, from which the patient could greatly benefit.

SUMMARY

(5) In some embodiments of the present disclosure, an implant system to treat glottic insufficiency is disclosed. The implant system includes a fixation frame comprising a first set of flanges at a first edge of the fixation frame and a second flange at a second edge of the fixation frame, wherein the fixation frame is configured to secure the implant system at an opening of the patient's thyroid cartilage, a port system disposed in the fixation frame and configured to receive, deliver, maintain, or remove a filler, and a flexible member, coupled to the fixation frame and in fluid or air communication with the port system, wherein based on an amount of the filler in the flexible member, the flexible member is configured to inflate in a direction to push against the patient's arytenoid cartilage so the arytenoid cartilage is rocked, adducted and/or inferomedially rotated.

(6) In some other embodiments of the present disclosure, a method to treat glottic insufficiency of a patient is disclosed. The method includes placing an implant system in an opening of the patient's thyroid cartilage, attaching a fixation plate and a fastener to a fixation frame of the implant system, wherein a thread goes through one or more holes of the fixation plate, securing the implant system on the thyroid cartilage with the fixation plate and the fastener, guiding an injector to a port system of the implant system, and injecting an amount of a filler to inflate a flexible member of the implant system so the flexible member pushes against the patient's arytenoid cartilage and rocks, adducts, and/or inferomedially rotates the patient's arytenoid cartilage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A and 1B illustrate various views of larynx anatomy.

(2) FIG. 1C illustrates various symptoms of glottic insufficiency.

(3) FIG. 2A illustrates a top view of a paraglottic space in which an implant system **200** can be placed to treat glottic insufficiency, in accordance with some embodiments of the present disclosure.

(4) FIG. 2B illustrates an example implant system **200** to be placed in a paraglottic space through an opening of thyroid cartilage **203**, in accordance with some embodiments of the present disclosure.

- (5) FIG. 2C illustrates an example implant system secured at an opening of a patient's thyroid cartilage and example approaches to rock, adduct, and/or inferomedially rotate the patient's arytenoid cartilage, in accordance with some embodiments of the present disclosure.
- (6) FIG. 2D illustrates an example embodiment of a flexible member of an implant system to treat glottic insufficiency, in accordance with some embodiments of the present disclosure.
- (7) FIG. 2E illustrates example configurations of a flow channel of a port system within an implant system disposed between a port chamber of the port system and a posterior member of flexible member, in accordance with some embodiments of the present disclosure.
- (8) FIG. 3A illustrates a top view of an example implant system secured at a thyroid cartilage to treat glottic insufficiency, in accordance with some embodiments of the present disclosure.
- (9) FIG. 3B illustrates an enlarged view of an example implant system configured to treat glottic insufficiency, in accordance with some embodiments of the present disclosure.
- (10) FIG. 3C illustrates a perspective view of an example implant system configured to treat glottic insufficiency, in accordance with some embodiments of the present disclosure.
- (11) FIG. 3D illustrates a cross sectional view of an example implant system configured to treat glottic insufficiency, in accordance with some embodiments of the present disclosure.
- (12) FIG. 3E illustrates an injection angle for an example implant system configured to treat glottic insufficiency, in accordance with some embodiments of the present disclosure.
- (13) FIG. 4 illustrates a flow chart of an example process to treat glottic insufficiency with an example implant system, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

- (14) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the Figures, can be arranged, substituted, combined, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.
- (15) The present disclosure describes apparatuses and methods to treat glottic insufficiency (due to neuromuscular incoordination or disordered interaction (cooperation) between the intrinsic muscles), and the poor closure of larynx inlet (opening) (due to uncoordinated bending of epiglottis and delayed or absence of laryngeal elevation). The disclosed treatments may improve the glottic closure and reduce the incidence of aspiration, thus preventing the aspiration's sequelae, such as aspiration pneumonia. Furthermore, the disclosed treatments may enable the postoperative adjustment of implants, and reduce the risk of postoperative airway compromise. As a result, the disclosed treatments may shorten the operation time during the surgical procedures. Perioperative complications, such as difficult intubation, can also be prevented through the implant adjustability. With a complete glottic closure, aspiration pneumonia can be effectively reduced or prevented after the surgical procedures.
- (16) In examples of the present disclosure, an implant system is designed to close the glottis by rocking, adducting, or inferomedially rotating the arytenoid cartilage. The amount of rock, adduction and/or inferomedially rotation is adjustable by changing the size of an implant, and the size of the implant can be changed by varying the amount of filler in the implant. The present disclosure may achieve the same effect as type I Medialization Thyroplasty (MT) and Arytenoid Adduction (AA), when performed together. Further, it may lengthen and tense the vocal fold by posterior expansion of the flexible member.
- (17) FIGS. 1A and 1B illustrate various views of larynx anatomy.
- (18) Drawing **110** shows the anatomy of a patient's larynx in an anterior view, with outer side **111** of the thyroid cartilage, or the thyroid cartilage lamina, exposed. Drawing **120** shows the same

patient's larynx in a posterior view, with inner side **121** of the thyroid cartilage exposed. Drawing **120** further shows the patient's vocal ligament **122**, which are enclosed within the patient's vocal folds (not shown in the drawing **120**). Drawing **130** shows the same patient's laryngeal cartilages (including a cross-sectional view **131** of the thyroid cartilage) in an angled lateral view. In examples of the present disclosure, the space surrounded by the laryngeal cartilages, including but not limited to thyroid cartilage, corniculate cartilage, arytenoid cartilage, cricoid cartilage, posterior cartilage illustrated in FIGS. **1A** and **1B**, as well as the tissues and organs connected to these cartilages, may be deemed a "paraglottic space." In other words, the paraglottic space may be a space bounded by the thyroid cartilage and the various surrounding membranes.

(19) FIG. **1C** illustrates various symptoms of glottic insufficiency.

(20) Drawing **140** illustrates an example healthy vocal fold having two lateralized vocal folds **141** and **142**. During normal breathing, swallowing, or speaking, vocal folds **141** and **142** open and close in unison. Before a person swallows food, the food or drink is first crushed and/or mixed into a pasty mass known as a bolus. During swallowing, the person's extrinsic and intrinsic muscles cooperate to prevent food or drink from entering the glottis. For example, the person's extrinsic muscles elevate the larynx and bend the epiglottis over the entrance to the glottis so that the bolus can glide across the epiglottis rather than falling into the larynx. While this movement is under way, the person's intrinsic muscles close the glottis. In conjunction with FIG. **1A**, should any food particles or liquids pass through the vestibule or vocal folds **141** and **142** or come in contact with trachea **143**, a cough reflex may be automatically triggered in a healthy person to prevent the material from falling more caudally. However, this cough reflex is desensitized in patients with neurological disorder, such as stroke, making it more important to completely close the glottic gap.

(21) Drawing **150** illustrates an example unhealthy vocal cord having a paralyzed vocal fold **151** unable to move to a fully lateralized/medialized position. In other words, paralyzed vocal fold **151** cannot be opened and closed in unison with the other vocal fold **152**, leaving an opening or gap in the glottis. Thus, during swallowing, the bolus may inadvertently slip into the glottis and subsequently into the trachea, bronchus, and lungs, which may lead to infection and pneumonia. Further, as shown in drawing **160**, for some of the stroke patients, due to impaired neurological stimulus, the trigger of epiglottis bending, the glottic closure **161**, and/or the laryngeal elevation **162** are delayed or absent, leading to incoordination of their extrinsic and intrinsic muscle movements. As a result, the patient may have difficulty in swallowing and may even choke as well.

(22) To treat a patient's glottic insufficiency and/or swallowing difficulty, drawing **170** shows that a surgeon or medical machinery may utilize a surgical device to create cartilage opening **172** in the patient's thyroid cartilage **171**. The surgical device may perform certain functions such as drilling, shaping, space expansion (e.g., "dissection"), and instrument/implant delivery. Afterwards, the surgeon or the medical machinery places a conventional implant system through cartilage opening **172** into the paraglottic space behind the patient's thyroid cartilage **171**.

(23) FIG. **2A** illustrates a top view of paraglottic space **201** in which implant system **200** can be placed to treat glottic insufficiency, in accordance with some embodiments of the present disclosure. The top of the figure corresponds to the anterior aspect of a patient's larynx, and the bottom of the figure corresponds to the posterior aspect of the patient's larynx. In FIG. **2A**, opening **203** of thyroid cartilage **210** is created to access paraglottic space **201** surrounded by thyroid cartilage **210** and arytenoid cartilages **221** and **222**. Opening **203** and thyroid cartilage **210** may correspond to opening **172** and thyroid cartilage **171** of FIG. **10**, respectively. Arytenoid cartilage **221** is coupled to healthy vocal fold **231**, and arytenoid cartilage **222** is coupled to paralyzed vocal fold **232**. Opening **203** may be created with a surgical device on the patient's thyroid cartilage lamina. The surgical device may perform certain functions such as, without limitation, drilling, shaping, and space expansion (e.g., dissection).

(24) In conjunction with FIG. **2A**, FIG. **2B** illustrates an example implant system **200** to be placed in paraglottic space **201** through opening **203**, in accordance with some embodiments of the present

disclosure. Implant system **200** includes fixation frame **240**, fixation block **250**, fastener **260**, port system **270**, and flexible member **280**. Implant system **200** may be secured to opening **203** with fixation frame **240**. Fixation frame **240** includes a first set of flanges **241**. Flanges **241** may be at a first edge of fixation frame **240**, and they are separated from each other by a distance corresponding to the thickness D of thyroid cartilage **210** to engage fixation frame **240** with opening **203** of thyroid cartilage **210**. This thickness D varies among patients. An example range of the thickness D is between 1 mm to 5 mm. Flanges **241** may or may not be substantially in parallel. Fixation frame **240** may include biocompatible materials, for example, titanium, peek, or silicone. In some other embodiments, fixation frame **240** may be elastic, so that it may be squeezed into and fit in various sizes of openings **203** of thyroid cartilage **210**. In addition, removing or reinstalling an elastic fixation frame **240** may be easier. Fixation frame **240** may further include a second flange **243** at a second edge of fixation frame **240**. Second flange **243** is configured to extend from the second edge of fixation frame **240**, away from the first edge adjacent to flanges **241**, to cover opening **203** of thyroid cartilage **210**.

(25) In some embodiments, fixation block **250** may include flange **257**. Fixation block **250** may be secured with fixation frame **240** after fixation frame **240** is placed into opening **203**. In some embodiments, fastener **260** is placed in fixation block **250** and is used to secure fixation block **250** to fixation frame **240**. Fastener **260** is configured to secure with second flange **243** of fixation frame **240**. Flange **257** and second flange **243** may be configured with a separation distance corresponding to the thickness D of thyroid cartilage **210** after fixation block **250** is secured with second flange **243** of fixation frame **240**. In addition, fixation block **250** may have wings, extending in anterior, cranial and caudal directions, providing a better halt for fixation block **250** and preventing fixation block **250** from falling into paraglottic space **201** through opening **203**.

(26) In conjunction with FIGS. 2A and 2B, FIG. 2C illustrates implant system **200** secured at opening **203** of thyroid cartilage **210** and example approaches to rock, adduct, and/or inferomedially rotate arytenoid cartilage, in accordance with some embodiments of the present disclosure. In some embodiments, by turning fastener **260**, fastener **260** may protrude medially from second flange **243** with a distance $D1$ and be physically in contact with flexible member **280**. As fastener **260** is turned further, fastener **260** may protrude further from second flange **243**, and the distance $D1$ increases. $D1$ may be less than 5 mm. In other embodiments, $D1$ may be between 0 mm to about 4 mm. By pushing against flexible member **280** in a medial or medioposterior direction, the change in the position of flexible member **280** creates a physical contact between flexible member **280** and arytenoid cartilage **222** and initiates a clockwise rotation of arytenoid cartilage **222**. This rotation leads to arytenoid cartilage **222** being rocked, adducted, and/or inferomedially rotated, which results in having the paralyzed vocal fold **232** fixed in its median position.

(27) In alternative embodiments, if vocal fold **231** is paralyzed, an implant system similar to implant system **200** may be placed from a right opening (not shown but from the same perspective of FIG. 2C) of thyroid cartilage **210** to fix vocal fold **231** in its median position. A flexible member of the implant system may be pushed in medial or medioposterior direction, initiate a counterclockwise rotation of arytenoid cartilage **221**, lead to arytenoid cartilage **221** being adducted and rotated inferomedially to fix vocal fold **231** in its median position.

(28) The head of fastener **260** may be circular, non-circular with multiple angles, star-shaped, or any shape to provide a sufficient halt between fastener **260** and surrounding material. Fastener **260** may be rotated with a screwdriver manually or may be rotated by electromagnetic force or remote control (e.g. Bluetooth) with an actuator integrated in implant system **200**.

(29) In addition to using fastener **260** to adjust arytenoid cartilage **222**, arytenoid cartilage **222** may also be adjusted with the AA surgical procedure. In some embodiments, fixation frame **240** provides a fixation point for one or more sutures used in the AA surgical procedure. One example of this fixation point may be a fastener. Fixation frame **240** may define opening **242**. One or more

sutures **290** can be inserted into opening **242** with one end **291** forming a loop to be in physical contact with the muscular process of the arytenoid cartilage **222** and/or lateral cricothyroid muscle and/or thyroarytenoid muscle to perform the AA surgical procedure. The other end of sutures **290** is secured on fastener **261** (e.g., screw) disposed adjacent to opening **242**. A surgeon may fasten fastener **261** to adjust the length and tension of sutures **290**. In response to the changes of the length and tension of sutures **290**, traction forces exerted on intrinsic laryngeal adductor muscles and/or arytenoid cartilage results in a rotation of arytenoid cartilage **222**, causing the change in the position in craniocaudal and/or mediolateral direction of a paralyzed vocal fold **232**.

(30) In addition to using fastener **260** and the AA surgical procedure, flexible member **280** may also be used to adjust arytenoid cartilage **222** and paralyzed vocal fold **232**. In conjunction with FIGS. 2A and 2B, in some embodiments, port system **270** is an apparatus configured to receive, deliver, maintain, or remove a filler (e.g., saline solution) to inflate or deflate flexible member **280**. In some embodiments, port system **270** includes port membrane **273**, port chamber **271**, and flow channel **272**. Port system **270** may be disposed in fixation frame **240**. To access port system **270**, a healthcare professional may use an injector, such as injector **295**, in direction **292** to puncture port membrane **273**. The filler may be injected into port system **270** after injector **295** reaches port chamber **271**. In some embodiments, the filler may also be removed from port system **270** with injector **295**. Port membrane **273** is flexible and is configured to automatically seal itself after injector **295** is removed. By sealing itself, the filler can then be maintained in port system **270** and flexible member **280**. Port membrane **273** may be made of silicone or other biocompatible materials of various thicknesses (>2 mm).

(31) Flexible member **280** is in fluid and/or air communication with the filler delivered, maintained, or removed through the port system **270**. In response to an increased pressure in port chamber **271** (e.g. when injector **295** is pushed), the filler flows into flexible member **280**, and flexible member **280** inflates. In response to a decreased pressure in port chamber **271** (e.g., when injector **295** is pulled), the filler leaves flexible member **280**, and flexible member **280** deflates. Flexible member **280** may have a shape that is suitable for introduction into paraglottic space **201**. The shape and size of flexible member **280** may be adjustable by adding/removing filler to/from flexible member **280** via port system **270** during operation (intraoperatively). After the operation (postoperatively), in response to possible compromises, such as airway compromises, intubation difficulties, and others, the size of flexible member **280** may be readjusted to prevent the occurrences of complications.

(32) FIG. 2D illustrates an example embodiment of flexible member **280** of implant system **200** to treat glottic insufficiency, in accordance with some embodiments of the present disclosure. In conjunction with FIG. 2C, in some embodiments, flexible member **280** includes anterior member **281** and posterior member **283**. Anterior member **281** is in fluid and/or air communication with fillers maintained in first port chamber **276** of port system **270**, and posterior member **283** is in fluid and/or air communication with fillers maintained in second port chamber **275** of port system **270**. First port chamber **276** and second port chamber **275** are separated by pressure valve **277**. Pressure valve **277** is configured to allow fillers to flow between first port chamber **276** and second port chamber **275** under a predetermined pressure. Inflations of anterior member **281** and posterior member **283** are controlled by the amount of fillers maintained in first chamber **276** and second chamber **275**, respectively. Inflation of anterior member **281** may cause flexible member **280** to push paralyzed vocal fold **232** back to a median position. Inflation of posterior member **283** may cause flexible member **280** to push against arytenoid cartilage **222**. The continued inflate of posterior member **283** would cause the adduction, rocking, and/or inferomedial rotation of arytenoid cartilage **222**. Posterior member **283** may include a posterior-medial tip **285**, which has a distance D2 of about 4 mm to about 15 mm measured perpendicularly from the medial side of the thyroid cartilage **210**. Flexible member **280** is made of biocompatible materials, such as silicon or biodegradable materials. To support a designated expansion direction, flexible member **280** may be

composed of a wall of varied stiffness (e.g., ≥ 20 Shore) and/or thickness (e.g., ≥ 2 mm).

(33) FIG. 2E illustrates example configurations of flow channel **279** of port system **270** disposed between port chamber **271** and posterior member **283** of flexible member **280**, in accordance with some embodiments of the present disclosure. In conjunction with FIG. 2D, flexible member **280** may include posterior member **283** but not anterior member **281**.

(34) As illustrated, fixation frame **240** may have different configurations, and posterior member **283** is in fluid and/or air communication with port chamber **271** via flow channel **279**. To prevent an injector configured to inject fillers into port system **270** from inadvertently puncturing flexible member **280** via flow channel **279**, in some embodiments, as shown in the top two figures of FIG. 2E, flow channel **279** is disposed in manners such that its entry **272** cannot be directly accessed by the injector. For example, entry **272** may be disposed adjacent to one end of port chamber **271**.

(35) In an alternative embodiment, as shown in the bottom figure of FIG. 2E, fixation frame **240** includes rigid plate structure **274** extended from one side of port chamber **271**, which is placed near entry **272**, to prevent the injector from directly accessing entry **272**.

(36) In conjunction with FIG. 2A, FIG. 3A illustrates a top view of an example implant system **300** secured at thyroid cartilage **210** to treat glottic insufficiency, in accordance with some embodiments of the present disclosure. In contrast to implant system **200** illustrated in FIGS. 2B-2D, injector **395** is configured to deliver a filler to posterior member **383** of flexible member **380** in direction **392**, causing posterior **383** to inflate approximately along direction **385** (e.g., inflate along the posterior-medial direction). Inserting injector **395** into implant system **300** in direction **392**, as opposed to direction **292** shown in FIG. 2C, allows a healthcare professional to more easily and precisely access port system administer. As posterior member **383** inflates, it eventually comes into physical contact with arytenoid cartilage **222** and pushes against arytenoid cartilage **222** to initiate clockwise rotation **387** of arytenoid cartilage **222**. While flexible member **380** and posterior member **383** may come in different shapes, they are configured to inflate in a particular manner, such as in a designated expansion direction (e.g., direction **385**). As discussed earlier, the wall of a flexible member may be varied in stiffness and/or thickness to support a designated expansion direction. In other words, instead of inflating posterior member **383** and flexible member **380** uniformly in all directions, the wall of flexible member **380** may be less stiff/thinner in the proximity of arytenoid cartilage **222** but more stiff/thicker elsewhere, so that posterior member **383** and flexible member **380** mainly inflate in direction **385** as the filler continues to be delivered.

(37) Additional details of implant system **300** are illustrated in FIGS. 3B-3E and described in the paragraphs below.

(38) FIG. 3B illustrates an enlarged view of an example implant system **300** configured to treat glottic insufficiency, in accordance with some embodiments of the present disclosure. Implant system **300** includes, but not limited to, thread **310**, fixation frame **320**, fixation plate **330**, screw **340**, port system **350** (shown in FIG. 3D), and flexible member **380**.

(39) In some embodiments, screw **340** is integrated with fixation plate **330**. Fixation plate **330** may include holes so that thread **310** can go through them. One end of thread **310** may be attached to base **325** of fixation frame **320**. Before screw **340** is fully engaged with base **325** and secures fixation plate **330** with base **325**, fixation plate **330** with the integrated screw **340** may move along thread **310**. With this illustrated configuration, thread **310** may help to ensure keeping fixation plate **330**, screw **340**, and fixation frame **320** together, especially during operation. Thread **310** may be removed after implant system **300** is secured at thyroid cartilage **210**.

(40) Fixation frame **320** is configured to secure implant system **300** at thyroid cartilage **210**. Fixation frame **320** includes a first set of flanges **321** and **321'**. Similar to flanges **241** shown in FIG. 2B, flanges **321** and **321'** may be at a first edge of fixation frame **320** and separated from each other with a distance corresponding to the thickness D of thyroid cartilage **210**. Flanges **321** and **321'** may or may not be substantially in parallel. Fixation frame **320** may include biocompatible materials, for example, titanium, peek, or silicone. Fixation frame **320** may further include a second

flange **323** at a second edge of fixation frame **320**. Second flange **323** is configured to extend from the second edge of fixation frame **320** to cover opening **203** of thyroid cartilage **210**.

(41) In some embodiments, fixation frame **320** includes base **325** to receive screw **340**. Base **325** is disposed adjacent to second flange **323** of fixation frame **320**. Screw **340** may be secured in base **325**. By securing screw **340** in base **325**, fixation plate **330** and second flange **323** may hold thyroid cartilage **210** to secure implant system **300** at thyroid cartilage **210**.

(42) In some embodiments, screw **340** may protrude in direction **360**, or medially, from second flange **323** with a certain distance and be physically in contact with flexible member **380**. As screw **340** protrudes further from second flange **323**, the distance of protrusion (not shown, but similar to **D1** of FIG. 2C) increases. In some embodiments, the protrusion distance may be less than 5 mm. In other embodiments, the protrusion distance may be between 0 mm to about 4 mm. By pushing against flexible member **380** in medial or medioposterior direction, the change in the position of flexible member **380** creates a physical contact between flexible member **380** and arytenoid cartilage **222** and initiates a clockwise rotation of arytenoid cartilage **222**. Similar to fastener **260** of FIG. 2C, this rotation leads to arytenoid cartilage **222** being rocked, adducted and/or rotated inferomedially, which results in having the paralyzed vocal fold **232** fixed in its median position.

(43) FIG. 3C illustrates a perspective view of an example implant system **300** configured to treat glottic insufficiency, in accordance with some embodiments of the present disclosure. In conjunction with FIGS. 3A and 3B, in some embodiments, fixation frame **320** further includes a pair of side arms **322** and anterior shovel **324**. Anterior shovel **324** may protrude anteriorly from port membrane **353**, and anterior shovel **324** may also be substantially perpendicular with any of side arms **322**. Anterior shovel **324** is disposed on one side of flange **321'** at an angle substantially greater than 90 degrees. Anterior shovel **324** may be configured to guide injector **395** to reach port membrane **353** of port system **350**. In some instances, anterior shovel **324** may also prevent injector **395** from puncturing the anterior aspect of a patient's larynx.

(44) Side arms **322** help to define a boundary around port membrane **353** of port system **350**. As injector **395** of FIG. 3A approaches port system **350**, injector **395** may hit side arms **322**, and the physical impact with side arms **322** may guide a surgeon to locate port membrane **353**.

Alternatively, side arms **322** may act as a visual guide for a surgeon to locate port system **350**, because side arms **322** are visible as bright hyperechoic lines in ultrasound images. In some embodiments, the thickness of side arms **322** is between about 0.5 mm to about 2 mm.

(45) FIG. 3D illustrates a cross sectional view of an example implant system **300**, in accordance with some embodiments of the present disclosure. In conjunction with FIGS. 3A, 3B, and 3C, in some embodiments, port system **350** is an apparatus configured to receive, deliver, maintain, or remove one or more fillers to inflate, maintain or deflate posterior member **383** of flexible member **380**. Port system **350** may be disposed in fixation frame **320** and may be adjacent to base **325**. In some embodiments, port system **350** includes port membrane **353**, port chamber **355**, and flow channel **357**. As injector **395** continues to move in direction **392**, the needle of injector **395** may puncture port membrane **353** and move into port chamber **355**. With the needle in port chamber **355**, injector **395** may then inject fillers into port chamber **355**. Posterior member **383** is in fluid and/or air communication with port chamber **355** via flow channel **357**. In some embodiments, flow channel **357** is disposed in manners such that its entry **359** cannot be directly accessed by the needle of injector **395**. For example, entry **359** may be disposed adjacent to one end of port chamber **355** to prevent injector **395** from inadvertently puncturing flexible member **380** via flow channel **357**. The interior wall of port system **350** may have an undulated surface to prevent sliding of port membrane **353** when injector **395** is inserted into or removed from the port system **350**.

(46) Injector **395** is also visible as a bright hyperechoic line in response to injector **395** being inserted in the same plane as the ultrasound beam of the ultrasound scanner. The alignment of injector **395** and the ultrasound beam may be achieved by a mechanical guide attached to a probe of the ultrasound scanner. In some embodiments, a surface treatment is applied on the injector to

increase the ultrasonic visualization. In some other embodiments, injector **395** may be an echogenic ultrasound needle. The echogenic ultrasound needle may include a polymer coating, which holds microbubbles and a dimpled shaft.

(47) FIG. **3E** illustrates an injection angle for an example implant system **300** configured to treat glottic insufficiency, in accordance with some embodiments of the present disclosure. Port membrane **353** of the port system **350** and surface **327** of fixation frame **320** may define an injection angle **329** for injector **395** to inject one or more fillers into the port system **350**. In some embodiments, injection angle **329** is about 0 degree to about 90 degrees. More specifically, injection angle **329** may be about 30 degrees to about 60 degrees (e.g., about 45 degrees). Orientation of the port membrane **353** and/or the port system **350** may be altered to adjust injection angle **329**. In some embodiments, port membrane **353** and/or the port system **350** may be oriented uniaxially, biaxially, or triaxially to achieve a proper injection angle **329**.

(48) FIG. **4** illustrates a flow chart of an example process **400** to treat glottic insufficiency with an example implant system **300**, in accordance with some embodiments of the present disclosure. In conjunction with FIGS. **3A**, **3B**, **3C**, and **3D**, example process **400** may include one or more operations, functions, or actions illustrated by one or more blocks, such as **402** to **410**. The various blocks may be combined into fewer blocks, divided into additional blocks, and/or eliminated depending on the desired implementation.

(49) Example process **400** may begin in block **402**. In block **402**, implant system **300** is placed in an opening **203** on a patient's thyroid cartilage **210**. As shown in FIG. **3A**, flanges **321** and **321'** are separated from each other with a distance corresponding to the thickness **D** of the thyroid cartilage so that flanges **321** and **321'** are able to engage with the thyroid cartilage. Block **402** may be followed by block **404**.

(50) In block **404**, fixation plate **330** and screw **340** are attached to fixation frame **320**. In some embodiments, screw **340** is integrated with fixation plate **330**. As discussed earlier, one end of thread **310** goes through one or more holes defined by fixation plate **330**. The other end of thread **310** is attached to base **325** of fixation frame **320**. Block **404** may be followed by block **406**.

(51) In block **406**, implant system **300** is secured on thyroid cartilage. One securing mechanism is to fasten screw **340** to lock fixation plate **330** to base **325**. In some embodiments, screw **340** may be fastened until the distance between fixation plate **330** and flange **323** substantially corresponds to the thickness **D** of the thyroid cartilage. In other embodiments, screw **340** may be turned further, so that screw **340** protrudes from base **325** and pushes against flexible member **380** in a medial or medioposterior direction. Block **406** may be followed by block **408**.

(52) In block **408**, injector **395** is guided to port system **350**. In some embodiments, side arms **322** are identified with ultrasonic waves to determine a boundary around the port system **350**. Injector **395** may be also identified with ultrasonic waves to adjust the path of injector **395** to approach the port membrane **353** of the port system **350** according to identified side arms **322**. In some embodiments, injector **395** is used to inject a filler into the port chamber **355** through the port membrane **353** of the port system **350**. Port chamber **355** is in fluid/air communication with posterior member **383** of flexible member **380** via flow channel **381**. In some embodiments, due to the injection angle of injector **395**, injector **395** may come in physical contact with side arms **322** and/or anterior shovel **324**. The feedback from the physical contact may cause a surgeon to adjust how injector **395** should be inserted and help to guide injector **395** to port membrane **353**. Block **408** may be followed by block **410**.

(53) In block **410**, certain amount of filler is injected by injector **395** so that the patient's arytenoid cartilage can be rocked, adducted, and/or rotated inferomedially. In some embodiments, posterior member **383** of flexible member **380** is adjusted based on the amount of the filler injected by injector **395**. Posterior member **383** may inflate in response to an increased amount of the injected filler. Similarly, posterior member **383** may deflate in response to a decreased amount of the injected filler. The inflation and deflation of posterior member **383** may adduct and inferomedially

rotate the arytenoid cartilage to place paralyzed vocal fold to a median position.

(54) Thus, apparatuses and methods for treating glottic insufficiency have been disclosed. Although one or more embodiments of the present disclosure have been described in some detail for clarity of understanding, it will be apparent that certain changes and modifications may be made within the scope of the claims. Accordingly, the described embodiments are to be considered as illustrative and not restrictive, and the scope of the claims is not to be limited to details given herein, but may be modified within the scope and equivalents of the claims. In the claims, elements and/or steps do not imply any particular order of operation, unless explicitly stated in the claims.

(55) Plural instances may be provided for components, operations or structures described herein as a single instance. Finally, boundaries between various components, operations and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of the disclosure(s). In general, structures and functionality presented as separate components in exemplary configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components. These and other variations, modifications, additions, and improvements may fall within the scope of the appended claims(s).

Claims

1. An implant system to treat glottic insufficiency of a patient, comprising: a fixation frame comprising a first set of flanges at a first edge of the fixation frame and a second flange at a second edge of the fixation frame, wherein the fixation frame is configured to secure the implant system at an opening of the patient's thyroid cartilage; a port system disposed in the fixation frame and configured to deliver, maintain or remove a filler; a flexible member, coupled to the fixation frame and in fluid or air communication with the port system, wherein based on an amount of the filler in the flexible member, the flexible member is configured to inflate in a direction to push against the patient's arytenoid cartilage so the arytenoid cartilage can be rocked, adducted and/or rotated inferomedially; a base disposed adjacent to the first edge of the fixation frame; a movable fixation plate on the base configured to secure with the base and hold the thyroid cartilage with the second flange; and a thread with an end attached to the base, wherein the movable fixation plate is movable along the thread.
2. The implant system of claim 1, wherein the movable fixation plate includes one or more holes for the thread to go through.
3. The implant system of claim 1, wherein a fastener is integrated with the movable fixation plate to secure the movable fixation plate to the base.
4. The implant system of claim 1, wherein the fixation frame further comprises a set of side arms forming a boundary around a port membrane of the port system.
5. The implant system of claim 4, wherein the side arms are made of a material responsive to an ultrasound and visible as bright hyperechoic lines in an ultrasound image.
6. The implant system of claim 4, wherein the side arms are disposed adjacent to the second edge of the fixation frame.
7. The implant system of claim 4, wherein the fixation frame further comprises a shovel protruding anteriorly from the port system.
8. The implant system of claim 7, wherein the shovel is configured to guide an injector to the port membrane.
9. The implant system of claim 1, wherein the port system comprises a port membrane, wherein the port membrane and a surface of the fixation frame defines an angle of about 0 degree to about 90 degrees to deliver the filler to the port system.
10. The implant system of claim 1, wherein the port system comprises: a port membrane; a port

chamber; and a flow channel, wherein the flow channel includes an entry disposed adjacent to an end of the port chamber.

11. The implant system of claim 1, wherein an interior wall of the port system has an undulated surface.

12. The implant system of claim 1, wherein the flexible member comprises a posterior member in fluid or air communication with the filler maintained in a port chamber defined by the port system, and in response to the filler flowing in the posterior member from the port chamber, the posterior member is inflated along a posterior-medial direction, causing the flexible member to push against the patient's arytenoids cartilage so the patient's arytenoids cartilage can be adducted, rocked, and/or inferomedially rotated.

13. The implant system of claim 12, wherein the flexible member is composed of a wall of varied stiffness and/or thickness to support expansion in the posterior-medial direction.

14. A method of using the implant system of claim 1 to treat glottic insufficiency of a patient, comprising: placing the implant system in the opening of the patient's thyroid cartilage; attaching the movable fixation plate and a fastener to the fixation frame of the implant system, wherein the thread goes through one or more holes of the movable fixation plate; securing the implant system on the patient's thyroid cartilage with the movable fixation plate and the fastener; guiding an injector to the port system of the implant system; and injecting an amount of the filler to inflate the flexible member of the implant system so the flexible member pushes against the patient's arytenoid cartilage and rocks, adducts, or inferomedially rotates the patient's arytenoid cartilage.

15. The method of claim 14, wherein the securing further comprises adjusting a distance between the movable fixation plate and the second flange of the implant system to substantially correspond to the thickness of the patient's thyroid cartilage.

16. The method of claim 14, wherein the guiding further comprises identifying a boundary of the port system based on a response of a pair of side arms around the port system to an ultrasonic wave.

17. The method of claim 16, wherein the guiding further comprises identifying the injector with the ultrasonic wave.

18. The method of claim 14, wherein the guiding further comprises adjusting movement of the injector based on feedback from physical contact with a pair of side arms around the port system, a shovel, or the pair of side arms and the shovel.

19. The method of claim 14, wherein the securing the implant system further comprises turning the fastener so the fastener protrudes from the fixation frame and pushes against the flexible member in a medial or medioposterior direction.
